

Final Project

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Summary

The FEM program was first expanded to three dimensions to handle a convection problem with exclusively essential boundary conditions. The program was then expanded to handle a convection boundary condition in one dimension. Finally, both parts were combined and a 3D problem with a 2D convection boundary condition was solved.

In Figure 2, the boundary conditions for the back of the cube was set at 100, while the sides were set to 20. The front had the convection boundary condition applied. The dimensions of the cube were 1x1x1, the convection coefficient was set to 100, and the conduction coefficient was set to 273.

As can be seen in Figure 3, the results seem to converge as the number of elements in the mesh increase.

Figures

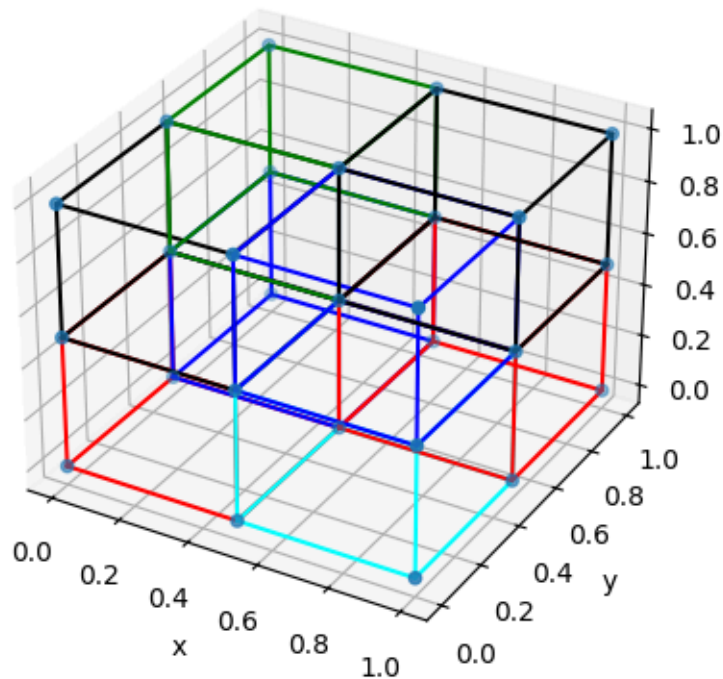


Figure 1: Final Mesh Visualization. An additional face was applied across the front face (like seen in Figure 2) to apply the convection boundary condition.

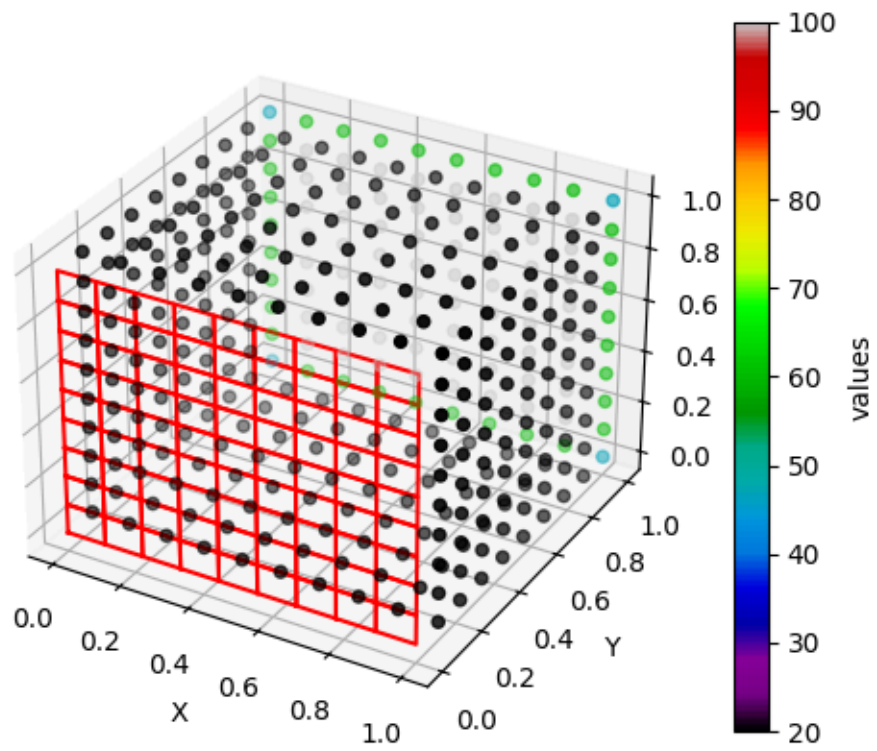


Figure 2: Final Boundry Conditions. The red wiring denotes the location of the convection boundry condition.

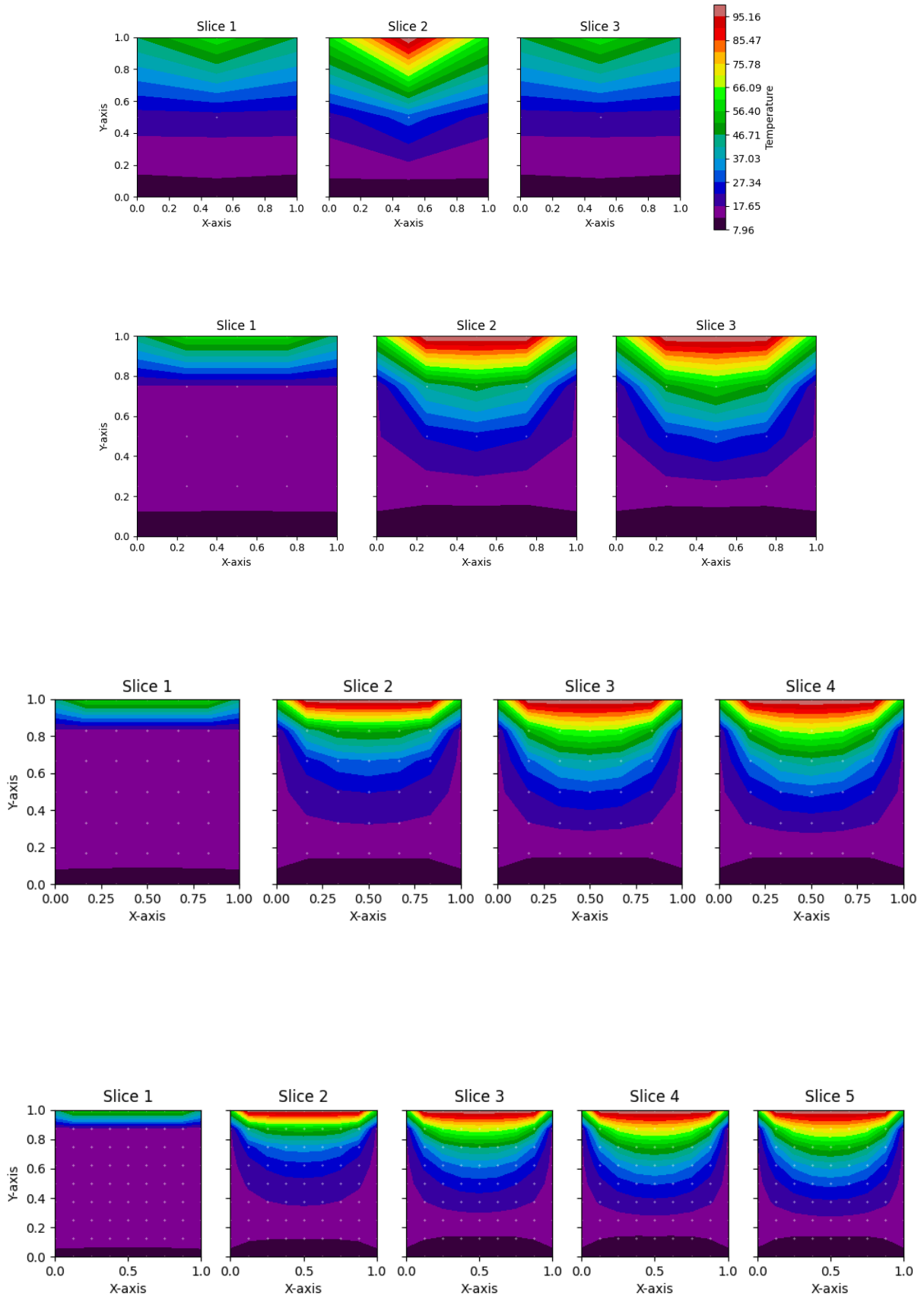


Figure 3: Final Mesh Visualization. Note that the white dots are the location of the nodes at each layer. Layers are symmetric so only half of the slices are shown except for the 3x3 mesh. Results are for 3, 5, 7, and 9 node deep meshes.